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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2025

BT2004-1 – UNIT OPERATIONS IN BIOPROCESS INDUSTRIES
(For BT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which among these is NOT a pressure measuring device?
A) Piezometer
B) U-tube manometer
C) Pressure gauge
D) Autoclave
2. If 'A' represents the area of the pipe and 'v' the flow velocity, then the discharge equation is given by
A) A/v
B) Av
C) Av^2
D) Av^2
3. In Bernoulli's equation, the term h_L represents
A) Energy loss due to height difference
B) Energy loss due to velocity difference
C) Energy loss due to friction
D) Energy loss due to temperature variations
4. The fluid whose density changes with temperature and pressure changes is called as
A) Incompressible fluid
B) Compressible fluid
C) Real fluid
D) Ideal fluid
5. For laminar flow,
A) $Re < 2100$
B) $Re = 5000$
C) $Re > 10000$
D) $Re = \text{infinity}$
6. Flow velocity through uniform pipe is
A) Maximum at the center of the pipe
B) Zero at the center of the pipe
C) Maximum at the walls of the pipe
D) Equal at the pipe wall and the center
7. Minor energy losses are NOT caused due to
A) Sudden expansion in pipe
B) Bend in pipe
C) Obstruction in pipe
D) Friction
8. According to continuity equation,
A) Mass in > mass out
B) Mass in < mass out
C) Mass in = mass out
D) Mass in $\neq$ mass out
9. Resistance to filtration is offered by
A) Pump, filter cake, and filter medium
B) Pump, filter medium, and filtrate
C) The feed, filter cake, and the filter medium
D) Slurry, filtrate, and cake
10. The three forces that are important in settling are:
A) Drag, buoyant, and gravitational force
B) Turbulent, drag, and gravitational force
C) Buoyant, turbulent, and drag force
D) Drag, buoyant, and centrifugal force
11. When settling is accelerated with the application of centrifugal force, then the process is called as
A) Filtration
B) Settling
C) Centrifugation
D) Mixing

BT2004-1

12. Filter cake whose porosity remains a constant is called as
 A) Incompressible cake B) Compressible cake
 C) Stiff cake D) Strong cake
13. Propellers are
 A) High speed impellers B) Low speed impellers
 C) Medium speed impellers D) Zero speed impellers
14. A bucket elevator is used for
 A) Measuring pressure drop B) Transportation of solids
 C) Transportation of liquids D) Transportation of gases
15. 200-mesh screen will have how many openings per linear inch?
 A) 2 B) 20
 C) 200 D) 2000
16. Which among these is NOT true?
 A) Size reduction improves dissolution rate B) Size reduction enhances binding strength
 C) Size reduction increases solubility D) Size reduction decreases surface area
17. The three parts of a venturimeter are
 A) Throat, orifice, plug B) Pipe, orifice plate, vena contracta
 C) Suction pipe, piston, delivery pipe D) Converging cone, throat, diverging cone
18. The ratio of throat diameter to orifice diameter is
 A) 0.25 B) 0.5
 C) 0.75 D) 1
19. Slip for a reciprocating pump is given by
 A) $Q_{th} - Q_{act}$ B) $Q_{act} - Q_{th}$
 C) $Q_{th} + Q_{act}$ D) Q_{th}/Q_{act}
20. Rotameter is a type of
 A) Variable area flow meter B) Variable pressure flow meter
 C) Variable head flowmeter D) Variable density flow meter

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) A physical quantity 'x' is known to depend on two other quantities 'y' and 'z.' Which method of dimensional analysis is suitable to derive the relation between these variables? Describe this method.	08	L*2	1	1
	b) Derive the expression for shear stress distribution for flow of viscous fluid through circular pipe. Depict the shear stress distribution profile.	08	L3	2	2
2.	a) What is U-tube manometer used for? Describe the construction of any one type of U-tube manometer and derive the manometric equation.	08	L3	1	2
	b) Differentiate between: i) Laminar and turbulent flow ii) Uniform and non-uniform flow iii) Steady and unsteady flow iv) Rotational and irrotational flow	08	L2	2	1
3.	a) State Newton's Law of viscosity. Classify fluids based on this law. Give one example for each of these fluid types.	08	L2	1	1
	b) Water is flowing through a pipe having diameter 300 mm and 200 mm at the bottom and upper end respectively. The intensity of pressure at the bottom end is 24.525 N/cm ² and at the upper end is 9.81 N/cm ² . Determine the difference in datum head if the rate of flow through pipe is 40 liter/s.	08	L3	2	2
Unit – II					
4.	a) What is flocculation? When is it used? State any two examples of flocculants.	04	L2	3	1

b)	What is?				
	i)	Constant pressure filtration			
	ii)	Constant rate filtration	04	L2	3 1
c)	Diagrammatically represent the parts of an agitated vessel.		08	L2	4 1
5.	a)	Illustrate rotary drum filter and describe its working mechanism.	08	L2	3 1
	b)	Define the following types of size reduction equipment:			
	i)	Crusher			
	ii)	Grinder			
	iii)	Ultrafine grinder			
	iv)	Cutting machine	08	L1	4 1
6.	a)	State any four basic assumptions of Kynch theory of sedimentation.	08	L1	3 1
	b)	Discuss the method of particle size measurement by sieve analysis.	08	L2	4 1

Unit – III

7.	a)	A single acting reciprocating pump running at 50 rpm delivers 0.01 m ³ /s of water. The diameter of the piston is 200 mm and the stroke length is 400 mm. The delivery and suction heads are 20 m and 5 m, respectively. Determine:			
	i)	Theoretical discharge of the pump			
	ii)	Co-efficient of discharge			
	iii)	Slip and % slip of the pump			
	iv)	Power required to drive the pump	08	L3	5 2
	b)	Describe the constructional features of venturimeter.	08	L2	5 1
8.	a)	Summarize the construction of centrifugal pump.	08	L2	5 1
	b)	List any four differences between orifice meter and venturimeter.	08	L2	5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
